

Greening While Governing the Economy: Do Green Central Banks Matter?

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04-05-2025

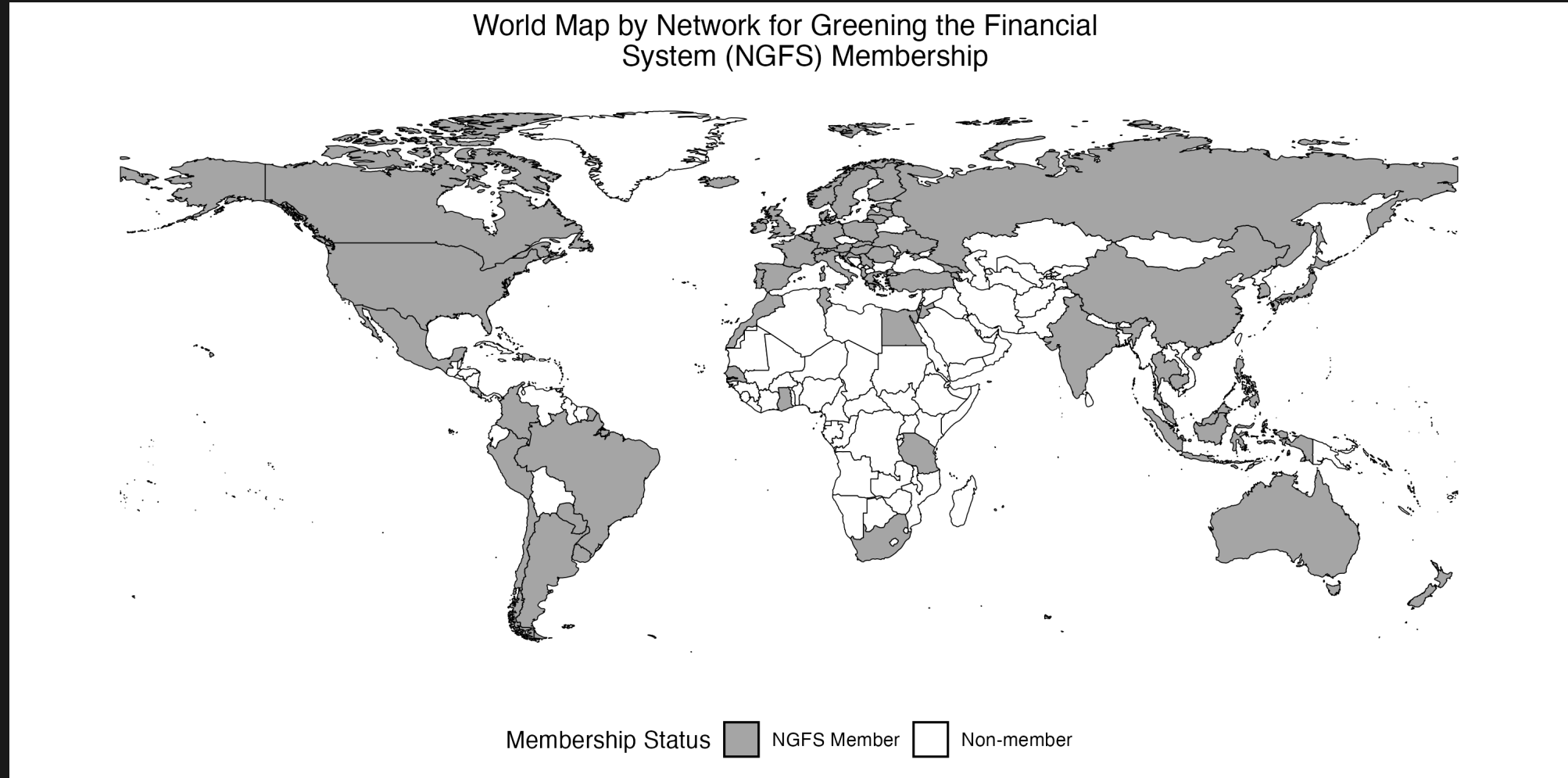
Central banks

- Central banks have historically worked to stabilize the economy and direct investments to the productive sector ([Evans 1995](#); [Wade 1990](#); [Zysman 1983](#))
- This includes when central banks shape industrial policy ([Amsden 2001](#); [Mazzucato 2021, 2024](#); [Rodrik 2008](#))
- Today, central banks are central to both economic and environmental policy-making as countries pursue their net-zero goals ([Battiston et al. 2021](#); [Keenan 2019](#); [Meckling 2021](#))

The economy and the environment

- Mixed evidence regarding the pursuit of either “brown growth” or “green growth” ([Ding 2022](#); [Kim and Thurbon 2015](#); [Rea and Frickel 2023](#); [Sabel and Victor 2024](#))
- Efforts still needed to recouple the economy and the environment ([Jorgenson and Clark 2012](#)), or re-embed the environment into the economy ([Polanyi 2001](#))
- To recouple the economy and the environment, central banks are expected to play a key role in decarbonizing the global economy ([Helleiner, DiLeo, and van 't Klooster 2025](#)). But do “green” central banks help reduce GHGs?

Novel dataset



Source: Prepared by the author

Data

- Data shared by the NGFS in March 2024
 - Central bank's country + year of membership
- Independent variable: NGFS member v. non-member
 - The French central bank joined the NGFS in 2017
 - France is coded as 1 from 2017 onward, after receiving “the treatment.” Coded as 0 otherwise
- Balanced panel data of 136 countries between 2001-2021
 - $N = 2,856$

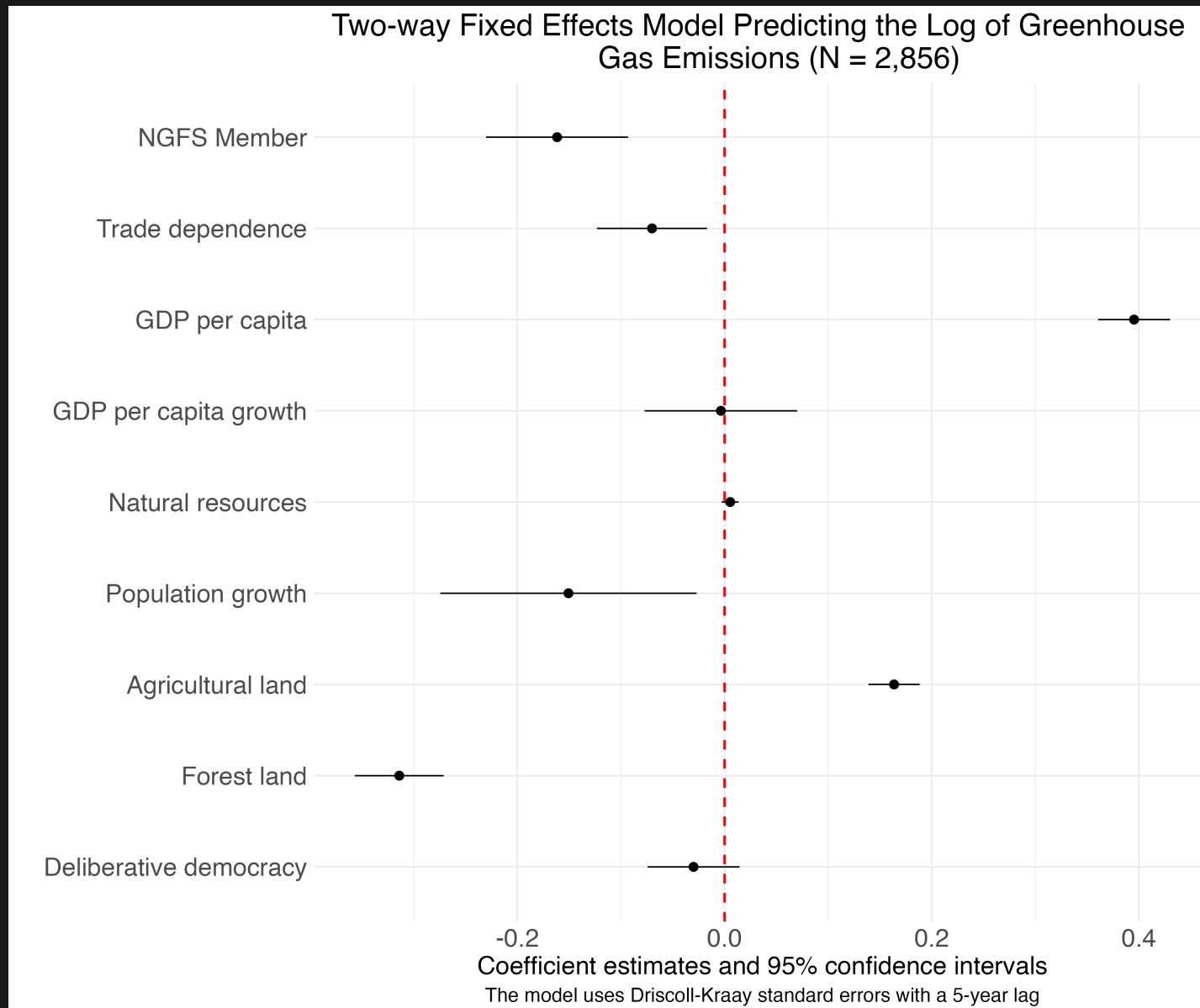
Dependent and control variables

- Dependent variable: Log of greenhouse gas emissions
 - Collected from the [Emissions Database for Global Atmospheric Research \(EDGAR\)](#)
 - Consistent harmonized data by country and sector
- Control variables have been collected from the World Bank and Varieties of Democracy (V-DEM)
 - Economic, environmental, and political covariates
 - Missing data for controls restricted the dataset to 2021

Descriptive statistics

Descriptive Statistics by Network for Greening the Financial System Membership Country x Years between 2001 and 2021 (N = 2,856)			
Variable	Member N = 195	Non-Member N = 2,661	Differences p-value
Greenhouse gas emissions	11.80 (1.83)	10.80 (1.85)	<0.001
Trade (% of GDP)	4.41 (0.59)	4.29 (0.50)	0.023
GDP per capita (constant 2015 US\$)	9.87 (1.06)	8.49 (1.45)	<0.001
GDP per capita growth (annual %)	3.93 (0.11)	3.95 (0.12)	0.008
Natural resources	-0.90 (2.48)	0.60 (2.17)	<0.001
Population growth (annual %)	2.52 (0.07)	2.59 (0.12)	<0.001
Agricultural land (% of land area)	5.81 (2.00)	6.11 (1.85)	0.092
Forest area (% of land area)	5.52 (1.83)	5.02 (2.27)	0.006
Deliberative democracy index (0-1)	0.60 (0.24)	0.44 (0.25)	<0.001
Region			
East and South Asia and Pacific	40 (21%)	401 (15%)	
Europe and Central Asia	97 (50%)	848 (32%)	
Latin America and Caribbean	23 (12%)	355 (13%)	
Middle East and Africa	29 (15%)	1,021 (38%)	
North America	6 (3.1%)	36 (1.4%)	
The values correspond to the mean for continuous variables and the proportion for categorical variables. The standard deviation is shown in parentheses. The data have been collected from the World Bank, the Network for Greening the Financial System, and the Varieties of Democracy (V-Dem) Project.			

Two-way fixed effects



Results

- Main finding: Joining the NGFS significantly reduces GHGs – about 10% less than non-members
- Controls behave in the expected direction
 - GDP per capita significantly increases GHGs, whereas population growth significantly decreases GHGs
- While the model is statistically causal, I theoretically read the results as an indicator that the NGFS's promotion of best practices seems to be moving in a sustainable direction

Discussion

- NGFS is based on voluntary membership
- Following Basel standards, it aims to standardize “brown” and “green” investments, estimate nature-related risks from a financial perspective, build capacity among members, and design transition plans from carbon-intensive ventures towards more sustainability-oriented investments
- A case of institutional design for common goods governance ([Ostrom 1990, 2005](#)), whereby stakeholders deliberate on collective mechanisms of financial and industrial policy that align economic and environmental factors

Robustness checks

- NGFS coefficient maintains the same direction using either country-level fixed effects or random effects
- Similar model behavior if other, though not so precise, controls are included, e.g., corruption index, state capacity, and population density
- Selection: Survival, Cox models reveal that GHG emissions and GDP PC significantly increase rather than decrease the probability of joining the NGFS
 - Next steps include trying to find and use IV for better estimation and other strategies

Thank You!

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Correlation Matrix

	Greenhouse gas emissions	Trade (% of GDP)	GDP per capita (constant 2015 US\$)	GDP per capita growth (annual %)	Natural resources	Population growth (annual %)	Agricultural land (% of land area)	Forest area (% of land area)	Deliberative democracy index (0-1)
Greenhouse gas emissions	1	.	.	.	.	.	.	.	.
Trade (% of GDP)	-.25	1	.	.	.	.	.	.	.
GDP per capita (constant 2015 US\$)	.36	.36	1	.	.	.	.	.	.
GDP per capita growth (annual %)	.02	.07	-.04	1	.	.	.	.	.
Natural resources	.04	-.30	-.51	-.01	1	.	.	.	.
Population growth (annual %)	-.14	-.15	-.32	-.11	.35	1	.	.	.
Agricultural land (% of land area)	.08	-.18	-.23	.08	-.04	-.11	1	.	.
Forest area (% of land area)	-.09	.04	.01	.04	-.11	-.20	-.21	1	.
Deliberative democracy index (0-1)	.12	.11	.60	-.00	-.50	-.31	.01	.26	1

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